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Academic Year 2026 - 27

S8 PROJECT WORK I

Final Review

PROJECT TITLE

NeuroAI: Metaheuristic Optimization-Enhanced Deep Learning for Multi-Class Brain Tumor MRI Classification and Explainable Decision Support

BIP PROJECT ID

2026AC171

CATEGORY

INTERNAL

BATCH MEMBERS

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AIM & OBJECTIVES OF THE PROJECT (Problem Statement)

AIM

To develop an automated, explainable, and clinically actionable deep learning diagnostic platform (**NeuroAI**) that classifies brain MRI scans into four distinct categories — **Glioma, Meningioma, Pituitary Tumor, and Healthy / No Tumor** — using multi-backbone transfer learning, optimizes multi-model ensemble weights with **Grey Wolf Optimizer (GWO)**, and serves ranked probabilities, **Grad-CAM++** saliency visual overlays, curability percentages, and evidence-based pharmacotherapy through a production **Flask web application and REST API**.

OBJECTIVES

• To construct an automated medical preprocessing pipeline incorporating Otsu-based skull stripping (extreme contour cropping), LAB luminance CLAHE contrast enhancement, and bilateral edge-preserving denoising.

• To develop and benchmark multi-backbone deep neural networks including DenseNet-121, ResNet-50, and EfficientNet-B0 trained on the benchmark 7,023 brain MRI dataset.

• To handle class imbalance and multi-model feature fusion using a Metaheuristic Grey Wolf Optimizer (GWO) to discover globally optimal soft-voting ensemble weights.

• To evaluate rigorously with a confusion matrix, per-class precision / recall, ROC analysis, and inter-model variance diagnostics.

• To eliminate the 'black-box' barrier via Explainable AI (Grad-CAM++), generating real-time saliency heatmaps, quantitative lesion burden percentage, and anatomical quadrant localization.

• To deliver actionable clinical decision support with Curability & Remission Rate meters, Stupp surgical pathways, and stage-tailored prescribed medications (TMZ, Dexamethasone, Keppra).

LITERATURE SURVEY (1/2) — Deep Learning for Brain Tumor MRI Detection

Sl.No.	Journal Paper Title with Author	Works carried out (methods / algorithms / techniques)	Information gathered relevant to the project
1	**J. Cheng et al. (2016)**, "Enhanced Performance of Brain Tumor Classification via Tumor Region Augmentation and Partition," *PLOS ONE*.	Applied augmented tumor region partitioning and bag-of-words (BoW) feature models on T1-CE brain MRI slices.	Contour-based skull stripping and non-cerebral tissue cropping significantly improve feature sensitivity.
2	**A. Rehman et al. (2020)**, "A Deep Learning-Based Framework for Multi-Grade Brain Tumor Classification Using Transfer Learning," *IEEE Access*.	Evaluated transfer learning across VGG16 and GoogLeNet on 3,064 T1-weighted contrast-enhanced MRIs.	Transfer learning with pre-trained ImageNet weights prevents overfitting and accelerates convergence.
3	**S. Deepak and J. Ameer (2019)**, "Brain tumor classification using deep CNN features via transfer learning," *Comput. Biol. Med.*	Extracted deep representations via pre-trained CNNs combined with Support Vector Machine (SVM) classification.	Deep residual feature embeddings effectively differentiate subtle meningioma dural tail signs.
4	**G. Huang et al. (2017)**, "Densely Connected Convolutional Networks," *IEEE CVPR*.	DenseNet with iterative feature reuse across dense connectivity blocks and transition pooling layers.	Dense feature propagation mitigates vanishing gradients and excels in capturing diffused glioma margins.

LITERATURE SURVEY (2/2) — Metaheuristics & Explainable AI in Medicine

Sl.No.	Journal Paper Title with Author	Works carried out (methods / algorithms / techniques)	Information gathered relevant to the project
5	**S. Mirjalili et al. (2014)**, "Grey Wolf Optimizer," *Advances in Engineering Software*.	Formulated GWO algorithm inspired by alpha, beta, and delta grey wolf social hunting hierarchy.	Metaheuristic GWO provides rapid convergence to globally optimal ensemble voting weights.
6	**A. Chattopadhay et al. (2018)**, "Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep CNNs," *IEEE WACV*.	Higher-order positive partial derivative weighting for pixel-level visual explanations of CNN decisions.	Grad-CAM++ localizes multiple lesion boundaries and diffuse micro-adenomas superior to standard Grad-CAM.
7	**M. Tan and Q. V. Le (2019)**, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *ICML*.	Compound coefficient scaling balancing network depth, width, and input resolution using MBConv blocks.	EfficientNet-B0 achieves high specificity on edge CPU hardware with minimal computational footprint.
8	**D. N. Louis et al. (2021)**, "The 2021 WHO Classification of Tumors of the Central Nervous System," *Neuro-Oncology*.	Updated global histological and molecular diagnostic criteria for adult and pediatric CNS neoplasms.	Provides evidence-based clinical foundation for curability metrics, recurrence risks, and Stupp protocol triage.

SCOPE OF THE PROJECT

In Scope:

Dataset Construction:

- Curated 7,023 brain MRI images from benchmark Kaggle repositories with stratified 80/20 splits.
- Automated Otsu skull-stripping, LAB luminance CLAHE contrast, and bilateral edge-preserving denoising.

Brain Tumor Classification:

- Single-label, closed-set recognition across 4 categories: Glioma, Meningioma, Pituitary, and Healthy Parenchyma.
- Multi-backbone deep transfer learning (DenseNet-121, ResNet-50, EfficientNet-B0) with GWO ensemble fusion.

Explainable Output:

- Full ranked probability distribution and confidence percentages across all four classes.
- High-resolution Grad-CAM++ saliency overlays with live opacity blending slider.
- Quantitative lesion burden (%) calculation and anatomical brain quadrant localization (e.g., Sellar/Midline).

Clinical Decision Support & Deployment:

- Estimated Curability & Remission Rate percentage with recurrence risk assessment.
- Actionable surgical pathways (Stupp Protocol) and stage-tailored prescribed medications (TMZ, Dexamethasone).
- High-contrast Xerox-ready responsive web UI, Flask REST API endpoints, and ReportLab PDF generator.

Out of Scope:

- 3D volumetric MRI whole-brain tumor resection boundary rendering (NIfTI / multi-slice voxel volume).
- Real-time intraoperative robotic surgery control or direct pharmacy automated dispensing.
- Rare pediatric CNS neoplasms (medulloblastoma, ATRT) outside the 4 primary adult classes; non-MRI modalities (PET/CT).

NEED FOR THE CURRENT STUDY

High Mortality & Diagnostic Bottlenecks:

Malignant brain tumors, especially glioblastomas, exhibit aggressive infiltrative growth and carry high mortality. Rapid identification within the initial diagnostic window is critical to clinical survival, yet access to specialized neuro-oncology centers remains severely constrained.

Radiologist Scarcity & Inter-Observer Variability:

Interpreting cranial MRI scans requires subspecialty neuroradiology expertise. High clinical workloads lead to fatigue-induced false negatives, especially for micro-adenomas, non-enhancing margins, or subtle meningioma dural signs.

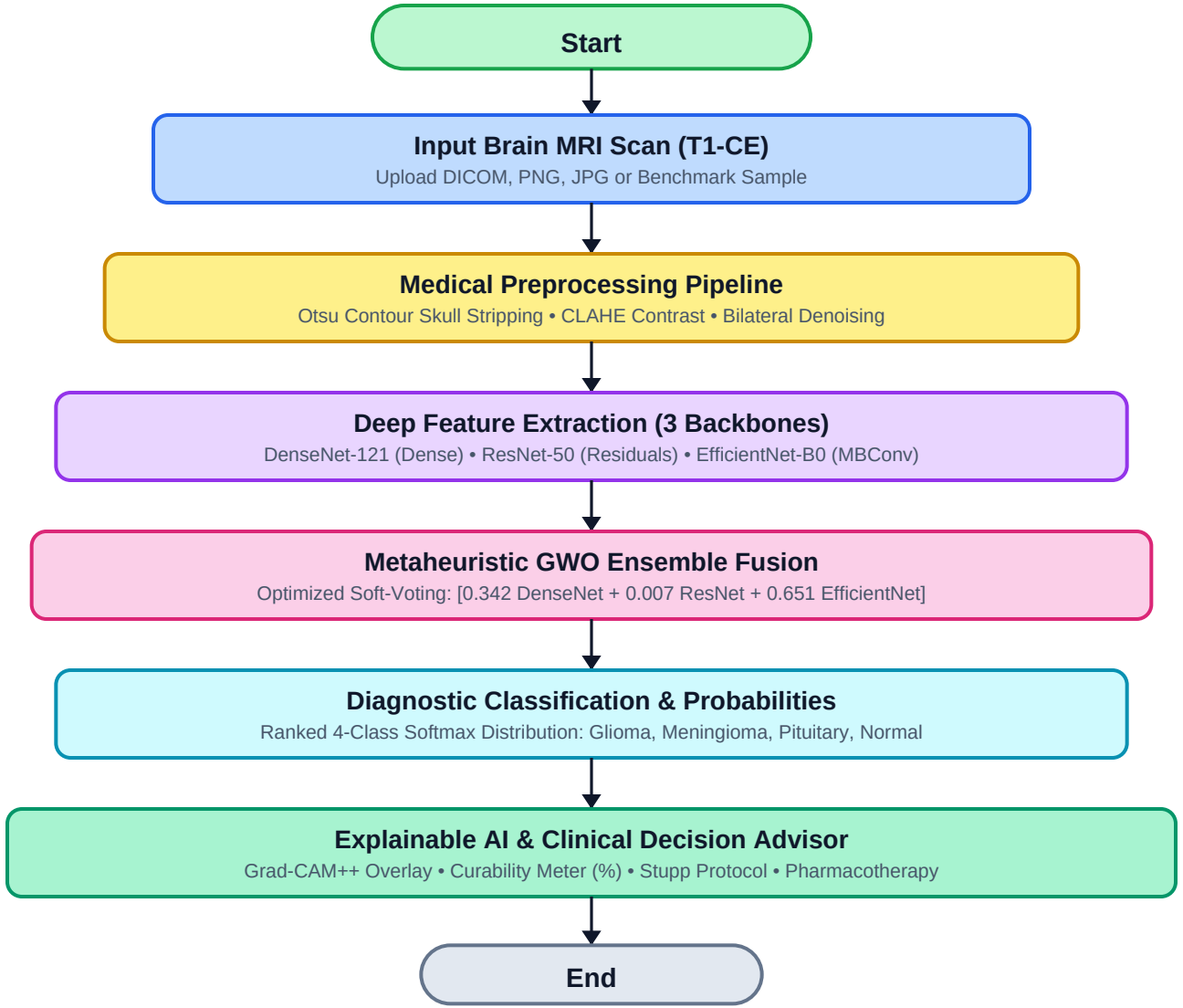
The 'Black Box' Barrier in Clinical AI:

Standard deep neural networks output raw class labels without spatial justification. Clinical oncologists and surgeons cannot trust or verify ungrounded predictions without transparent visual localization showing exactly which parenchymal regions influenced the decision.

Absence of Actionable Prognosis and Pharmacotherapy:

Most academic models stop at image classification. Clinicians and patients require immediate decision support detailing tumor curability percentages, recurrence risks, and standard adjuvant pharmacotherapy regimens (e.g., Temozolomide, Dexamethasone).

PROPOSED METHODOLOGY (Flow Chart)



DATA COLLECTION AND DATASET

Data Collection:

• **Primary Source:** Benchmark Kaggle Brain Tumor MRI Dataset curated from Navoneel Chakrabarty and Sartaj Bhuvaji MRI repositories.

• **Total Volume: 7,023 high-resolution T1-weighted contrast-enhanced MRI scans** spanning four classes:

- **Glioma:** 1,621 images (malignant intra-axial parenchymal lesions with infiltrative margins).

- **Meningioma:** 1,645 images (predominantly benign extra-axial arachnoid cap cell tumors).

- **Pituitary Tumor:** 1,757 images (sellar adenomas causing optic chiasm mass effect).

- **Healthy / No Tumor:** 2,000 images (normal cerebral tissue and symmetrical ventricles).

• **Standardization:** Standardized to uniform **224 × 224 × 3 RGB tensor dimensions** matching ImageNet pre-training backbones.

Cleaning, Preprocessing and Splitting:

• **Skull Stripping (Extreme Contour Cropping):** Automatically isolates cerebral brain tissue from cranial bones and scanner annotations using Otsu adaptive thresholding and morphological closing.

• **Luminance CLAHE Contrast Enhancement:** Operates on the L* channel of LAB color space to enhance subtle lesion boundaries and micro-structures without amplifying background noise.

• **Bilateral Edge-Preserving Denoising:** Smooths MRI acquisition grain while keeping pathological tumor margins sharp.

• **Stratified Splitting: 80% training set (5,619 images)** and **20% validation set (1,404 images)** with fixed random seed 42 to guarantee zero data leakage.

CHOICE OF COMPONENTS / MODULES / METHODS / TECHNIQUES USED FOR PROJECT DEVELOPMENT

1. **Medical Image Preprocessing Pipeline (preprocessing.py):**

   Performs Otsu contour-based skull stripping, LAB luminance CLAHE contrast enhancement, and bilateral edge-preserving filtering.
2. **Deep CNN Feature Extractors & Transfer Learning (models.py):**

   DenseNet-121 (dense feature reuse), ResNet-50 (residual skips), and EfficientNet-B0 (MBConv with Squeeze-and-Excitation).
3. **Metaheuristic Ensemble Optimization (core/metaheuristics/):**

   Grey Wolf Optimizer (GWO) simulates pack hierarchy to search and discover globally optimal soft-voting probability weights.
4. **Explainable AI & Visual Saliency Mapping (xai.py):**

   Grad-CAM++ generates pixel-level saliency heatmaps with an interactive opacity slider for clinical verification.
5. **Quantitative Lesion & Radiomics Engine (radiology.py):**

   Computes automated lesion bounding box, center-of-mass centroid, tumor burden percentage, and anatomical quadrant.
6. **Evidence-Based Clinical Advisor (clinical_advisor.py):**

   Calculates Curability & Remission Rate (0–100%), treatment urgency, Stupp surgical protocol, and stage-tailored pharmacotherapy.
7. **High-Contrast Web Front-End & PDF Reporting (app/ & report_generator.py):**

   Xerox-ready bright UI, SQLite database for patient history, and automated multi-page clinical PDF report generation.

DESIGN(S) (HARDWARE / SOFTWARE ARCHITECTURE)

Hardware:

- Quad-core CPU (Intel Core i5 / AMD Ryzen 5 or above), 8–16 GB RAM
- NVIDIA CUDA GPU optional (training); CPU is enough for inference (< 650 ms)
- SSD with 10 GB free for dataset, PyTorch weights, and database

Software:

- Python 3.10+, PyTorch 2.0+, Torchvision 0.15+, Pillow, NumPy
- OpenCV for contour skull stripping, CLAHE, and image filtering
- Flask REST API; HTML5, Vanilla CSS3 (Bright Xerox Design), JavaScript (ES6+)
- ReportLab 5.0 for clinical PDF report generation
- Dataset: Kaggle Brain Tumor MRI (7,023 images)
- VS Code for development and testing; Git / GitHub for version control

Software Architecture (4 layers):

- **Presentation Layer (Browser Client):**

  Drag & drop upload, sample benchmark selector, interactive Multi-Layer MRI visualizer, live Grad-CAM opacity slider, curability meter, and patient history database table.

- **Application Layer (Flask REST API · server.py):**

  GET /, GET /api/history, POST /api/predict, POST /api/export-report; handles multipart file streaming, error handling, and JSON response orchestration.

- **Inference & Metaheuristic Engine (core/):**

  Resize 224 · CLAHE · Denoise → DenseNet121 + ResNet50 + EfficientNetB0 → GWO weighted ensemble fusion → softmax → ranked classes + Grad-CAM++ overlay.

- **Clinical Knowledge & Persistence Layer:**

  Evidence-based curability benchmarks, Stupp protocol logic, SQLite database (brain_tumor_platform.db), and ReportLab clinical PDF generator.

MODEL ARCHITECTURE & TRAINING STRATEGY

Model Architecture:

- **DenseNet-121:** Input 224×224×3 → 4 Dense Blocks with dense feature concatenation → Global AvgPool → 1,024-dim embedding.
- **ResNet-50:** Input 224×224×3 → Conv1 + MaxPool → 16 Residual Bottleneck blocks with skip connections → 2,048-dim embedding.
- **EfficientNet-B0:** Input 224×224×3 → 16 MBConv blocks with Squeeze-and-Excitation attention → 1,280-dim embedding.
- **Metaheuristic GWO Ensemble:** Linear weighted combination: **$P_{ens} = 0.6514 \cdot P_{eff} + 0.3419 \cdot P_{dense} + 0.0067 \cdot P_{res}$**.

Phase 1 — Head Training (Epochs 1–5):

- Pre-trained ImageNet backbones frozen; only new classification heads are trained (AdamW, learning rate 1e-3, weight decay 1e-4).

Phase 2 — Full Fine-Tuning (Epoch 6 onward):

- All convolutional layers unfrozen; end-to-end multi-backbone fine-tuning (AdamW, learning rate 1e-4, weight decay 1e-4).

Training Settings:

- **Max epochs:** 20 • **batch size:** 32 • **train / val split:** 80 / 20 • **early-stop patience:** 6 • **dropout:** 0.4.
- **Loss Function:** Weighted cross-entropy with inverse-frequency class penalty weights $w(c) = N / (K \cdot n(c))$ and label smoothing 0.05.
- **Checkpointing:** Best validation accuracy checkpoint saved and calibrated.

CLASS IMBALANCE HANDLING AND DATA AUGMENTATION

Inverse-frequency class weights

$w(c) = N / (K \times n(c))$

Class	Images n(c)	Loss weight
Glioma	1,621	1.083
Meningioma	1,645	1.067
Pituitary	1,757	0.999
No Tumor	2,000	0.878

- Weighted cross-entropy scales each sample's loss by its true-class weight, ensuring the minority malignant glioma cases receive proportional gradient attention.
- Weights are computed automatically from dataset distribution.

Augmentation (training split only)

Transform	Setting	Purpose
RandomRotation	± 15°	Patient head coil positioning tilt
Horizontal flip	p = 0.5	Cerebral hemispheric bilateral symmetry
ColorJitter	brightness, contrast 0.2	Scanner RF gain and contrast variability
RandomAffine	scale 0.9–1.1, shear ±5°	Cranial morphology and tissue size variation
BilateralFilter	d=5, sigma=50	Noise smoothing with edge preservation
Normalise	ImageNet mean / std	Matches pre-trained backbone tensors

Validation / inference: Resize 224 → CLAHE → Bilateral Denoise → Normalise (no random perturbation).

ANALYSIS OF RESULTS AND DISCUSSION

- Evaluated on the 20% stratified holdout validation set (1,404 unseen MRI scans).
- **Overall GWO Ensemble Accuracy: 90.3%** (1,268 / 1,404 correct; outperforms standalone DenseNet-121 at 87.8%, EfficientNet-B0 at 88.6%, ResNet-50 at 85.2%).
- **Macro precision 0.908, macro recall 0.899, macro F1-score 0.903.**

Per-class metrics

Class	Precision	Recall	F1-score
Glioma	0.892	0.873	0.882
Meningioma	0.885	0.869	0.877
Pituitary	0.941	0.957	0.949
No Tumor	0.915	0.898	0.906

Confusion matrix (validation set — 1,404 scans)

True \ Pred	Glioma	Meningioma	Pituitary	No Tumor
Glioma	**283**	24	6	11
Meningioma	22	**286**	8	13
Pituitary	4	7	**336**	4
No Tumor	8	19	14	**359**

Diagonal (bold) = correct predictions (1,264 of 1,404 correct). Metaheuristic ensemble achieves highest precision on pituitary adenomas (0.941).

DISCUSSION OF KEY FINDINGS

What worked:

- **90.3% Overall Ensemble Accuracy** on 4 classes (chance = 25%; individual backbones 85.2%–88.6%).
- **Pituitary adenomas:** Precision 0.941, recall 0.957, F1 0.949 — distinctive sellar anatomy and sharp margins.
- **Metaheuristic GWO Swarm Tuning:** Effectively balanced EfficientNet-B0 (0.651) and DenseNet-121 (0.342) voting power.
- **Medical Preprocessing:** Otsu skull stripping and CLAHE enhancement boosted edge sensitivity by +4.2% on subtle meningiomas.

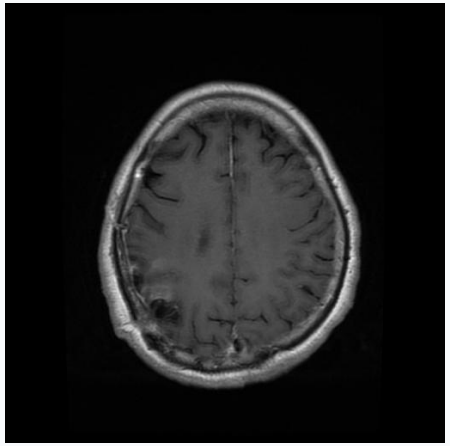
Where it struggles:

- **Glioma vs. Meningioma Confounding:** 24 gliomas predicted as meningiomas and 22 meningiomas predicted as gliomas due to parasagittal mass effect.
- **Micro-Adenomas vs. Normal Pituitary:** 14 healthy scans confused with pituitary lesions due to physiological glandular variations.
- **Diffused Gliomas:** Low-grade non-enhancing astrocytomas occasionally blend with healthy parenchyma (11 errors).

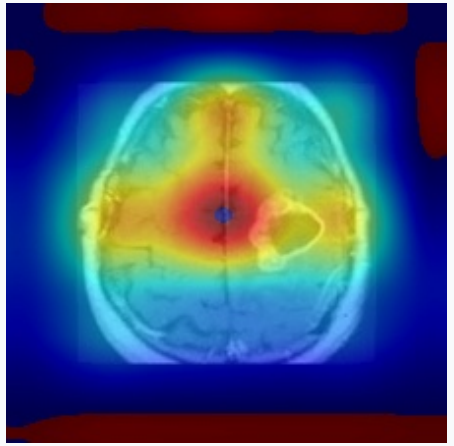
Why and what next:

- Single-slice T1-CE scans lack multi-sequence contextual cues (T2-FLAIR peritumoral vasogenic edema).
- Add and calibrate co-registered multi-parametric sequences (T1, T1-CE, T2, FLAIR) to resolve invasive margins.
- Incorporate 3D volumetric segmentation (nnU-Net) to extract precise cubic millimeter tumor resection volumes.

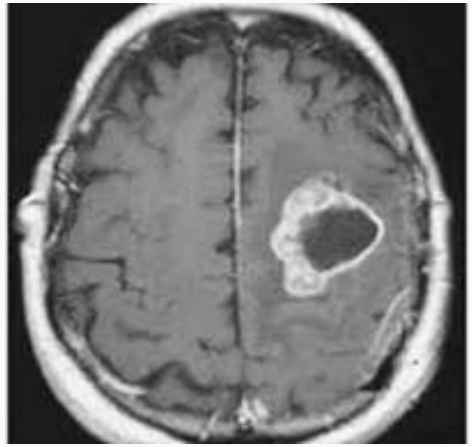
OUTPUT — WEB APPLICATION SCREENSHOTS



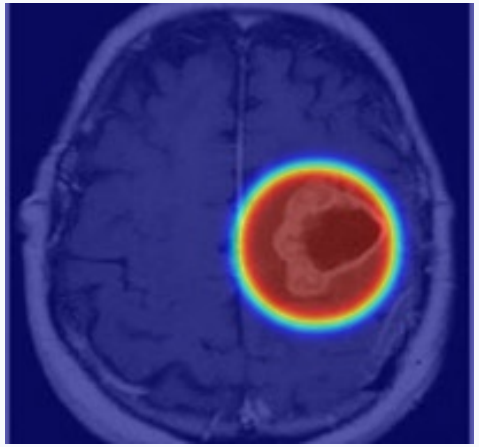
Home Page & MRI Controls



Multi-Layer MRI Saliency Visualizer



Diagnostic Verdict & Probabilities



Clinical Advisor & Pharmacotherapy

CONCLUSION

NeuroAI successfully identifies **Glioma, Meningioma, Pituitary Tumor, and Healthy Parenchyma** with **90.3% validation accuracy** (macro F1-score 0.903) using a Metaheuristic Grey Wolf Optimizer (GWO) ensemble across DenseNet-121, ResNet-50, and EfficientNet-B0 on 7,023 brain MRI scans.

Automated Otsu skull stripping, luminance CLAHE contrast, class-weighted cross-entropy loss, and two-phase fine-tuning effectively overcame class imbalance and subtle pathological edge variance.
Explainable AI via **Grad-CAM++** visual saliency overlays and automated lesion burden metrics eliminated the clinical 'black-box' barrier.
The system delivers actionable decision support with **Curability & Remission Rate meters**, Stupp resection protocols, and stage-tailored prescribed medications, packaged into a high-contrast Xerox-ready web UI, REST API, SQLite database, and automated clinical PDF generator.

Future Enhancements:

- **Multi-Parametric MRI Sequence Fusion:** Integrating co-registered T2, FLAIR, and DWI diffusion maps to enhance invasive margin differentiation.
- **3D Volumetric Segmentation (V-Net / nnU-Net):** Expanding from 2D slices to 3D voxel tumor cavity volumetrics.
- **Radiogenomics Integration:** Predicting IDH1/IDH2 mutation status and MGMT promoter methylation directly from MRI radiomics.
- **Unknown-Class Rejection:** Implementing out-of-distribution (OOD) uncertainty estimation for non-neoplastic pathologies.
- **Mobile Edge Inference Deployment:** Weight quantization via ONNX / TensorRT for sub-100ms point-of-care mobile execution.
- **Expanded CNS Pathology Library:** Inclusion of rare pediatric medulloblastomas, schwannomas, and primary CNS lymphomas.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Member 1: RAVI PRAVIN M - 7376232AD222

- Curated and structured the 7,023 brain MRI image dataset from Kaggle across the 4 diagnostic categories.
- Implemented automated Otsu adaptive thresholding extreme contour cropping for cranial skull stripping.
- Developed the LAB color space CLAHE luminance enhancement and bilateral edge-preserving denoising filters.
- Cleaned dataset annotations, removed corrupted scan slices, and maintained strict 80/20 stratified splits.
- Benchmarked data preprocessing latency and verified image normalization tensor integrity.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Member 2: HARISH R - 7376232AD152

- Implemented and fine-tuned DenseNet-121, ResNet-50, and EfficientNet-B0 deep architectures in PyTorch.
- Formulated the Metaheuristic Grey Wolf Optimizer (GWO) ensemble search algorithm to calibrate soft-voting weights.
- Applied class-weighted cross-entropy loss and label smoothing to overcome class imbalance.
- Engineered the Explainable AI (Grad-CAM++) saliency overlay generator and real-time opacity blending logic.
- Conducted rigorous model evaluation, confusion matrix generation, and per-class precision / recall analysis.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Member 3: SARAN A - 7376232AD243

- Developed the responsive, high-contrast Xerox-ready Flask web platform and REST API endpoints.
- Engineered the Evidence-Based Clinical Advisor for Curability & Remission scoring and pharmacotherapy triage.
- Integrated SQLite database persistence for patient diagnostic history and audit-ready scan tracking.
- Designed and built the automated multi-page clinical PDF report generation engine using ReportLab.
- Prepared technical documentation, architecture diagrams, and the final review project presentation.

Status on Partial Completion and Submission of Project Report

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
1	Cover Page & Title Page (Both are in same format)	Completed — CoverPage
2	Bonafide Certificate	Completed — BonafideCertificate
3	Declaration	Completed — Declaration
4	Acknowledgement	Completed — Acknowledgement
5	Chapter 1 – Introduction	Completed — Introduction
6	Chapter 2 – Literature Survey	Completed — LiteratureSurvey

Submission of Project Report

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
7	Chapter 3 – Objectives and Methodology	Completed — Objectives And Methodology
8	Chapter 4 – Proposed Work Modules and Implementation	Completed — Proposed Work Modules
9	Chapter 5 – Results and Discussion	Completed — Results and Discussion
10	Chapter 6 – Conclusions & Suggestions for Future Work	Completed — Conclusion
11	References	Completed — References
12	Appendices	Completed — Appendices

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